

Assignment 2 — Batch Growth Kinetics

Q1. Growth from experimental data

A microorganism is cultivated in a batch reactor. During exponential growth, the following biomass concentrations are measured:

Time (h)	Biomass X (g/L)
2	0.80
4	1.60
6	3.20

- Calculate the specific growth rate, μ , between 2 and 6 h.
- Calculate the biomass doubling time.
- Would you expect the same μ to apply throughout the entire batch cultivation? Explain.

Q2. Interpreting a batch-growth profile

A batch culture initially contains **20 g/L glucose**. Biomass increases rapidly between 2 and 8 h, while glucose decreases. At approximately 10 h, glucose approaches zero and biomass concentration stops increasing.

Answer the following:

- What growth phase is most likely occurring between 2 and 8 h?
- Why does biomass stop increasing at approximately 10 h?
- Does the cessation of net biomass accumulation necessarily mean that all cellular metabolic activity has stopped? Explain.

Q3. Biomass yield

A batch fermentation starts with:

- Biomass = 0.5 g/L
- Glucose = 25 g/L

At the end of cultivation:

- Biomass = 10.5 g/L
- Glucose = 5 g/L

- Calculate the observed biomass yield coefficient, $Y_{X/S}$.
- Explain biologically what this value means.
- Why should the initial biomass not simply be included as biomass produced from the substrate during this batch?

Q4. Specific substrate-consumption rate

During exponential growth:

$$\mu = 0.30 \text{ h}^{-1}$$

and

$$Y_{X/S} = 0.60 \text{ g}_X/\text{g}_S$$

Assuming substrate consumption is associated predominantly with biomass formation:

- a. Estimate the specific substrate-consumption rate q_S .
- b. Explain the difference between q_S and the overall substrate-consumption rate of the reactor.

Q5. Product formation

Two microbial products show the following behavior:

Product A: production rate increases strongly during exponential growth and falls when growth stops.

Product B: little product accumulates during exponential growth, but substantial production occurs after biomass growth has slowed.

- a. Which product behaves predominantly as a growth-associated product?
- b. Which behaves predominantly as a non-growth-associated product?
- c. Why should these categories be regarded as useful approximations rather than absolute biological classes?

Q6. Integrated interpretation

A batch fermentation shows the following sequence:

Substrate decreases → biomass increases → growth rate eventually decreases → product continues accumulating after biomass reaches its maximum.

A student concludes:

“The cells stop metabolizing once biomass stops increasing.”

Evaluate this conclusion and propose a more accurate interpretation of the observed profiles.